ASSESSMENT

Weekly homework — problems

- 1. PCA by hand: for C = [[2,1],[1,2]], find eigenvalues/eigenvectors, explained variance, and the 1-D projection of $x = [2,0]^T$.
- 2. A standardized 8-feature audio set has explained-variance ratios [0.38,0.24,0.15,0.09,0.06,0.04,0.025,0.015]. Choose k for ≥90% retention and explain one limitation of this rule.
- 3. SVM: explain the effect of increasing C and increasing γ (RBF). Give one underfit and one overfit scenario.
- 4. Compute Gini gain for parent [20,20] split into [18,2] and [2,18]. Compare with a split into [12,8] and [8,12].
- 5. Explain why a depth-20 audio tree can fail on a new recording room even with 100% training accuracy. Name two regularization choices.
- 6. Random forest: explain bootstrap sampling, random feature subsets, OOB prediction, and why correlated trees limit the benefit of averaging.
- 7. In the Lecture 23 notebook, compare scaled RBF SVM with and without PCA using 5-fold CV. Report mean±std and interpret.
- 8. In the Lecture 24 notebook, compare MDI and permutation importance after adding a noisy duplicate of spectral centroid. Explain any change.

ASSESSMENT

Weekly quiz — 10 minutes

1. PCA chooses directions that maximize:

A. class separation B. variance C. margin D. entropy

2. Why must scaling/PCA be inside a CV pipeline?

A. speed B. avoid label imbalance C. avoid leakage D. make trees work

3. Larger SVM C generally means:

A. cheaper violations B. stronger penalty on violations C. wider RBF D. fewer features

4. Which model does NOT require feature standardization?

A. PCA B. RBF SVM C. decision tree D. linear SVM

5. Main purpose of random feature subsets in a forest:

A. normalize data B. de-correlate trees C. eliminate bootstrapping D. guarantee causality

6. Short answer: Why can a highly important feature in a random forest still be non-causal?

Scoring

1 point each. For #6, accept "predictive association can arise from correlation/confounding; the model was not a causal experiment."

INSTRUCTOR APPENDIX

Homework + Quiz Solutions

Hide or remove these slides before distributing a student-facing deck.

INSTRUCTOR APPENDIX

HW 1–2 solutions

1. PCA by hand

$\lambda_1=3$ with $v_1=(1/\sqrt{2})[1,1]^T$; $\lambda_2=1$ with $v_2=(1/\sqrt{2})[1,-1]^T$. PC1 retains 75%. For $x=[2,0]^T$, score on PC1 is $v_1^T x=\sqrt{2}$ (assuming x is already centered in this coordinate system).

2. Explained variance

Cumulative: 0.38, 0.62, 0.77, 0.86, 0.92... Therefore k=5 for ≥90%. Limitation: variance retention is unsupervised; a low-variance direction can still carry class information.

INSTRUCTOR APPENDIX

HW 3–4 solutions

3. SVM hyperparameters

Increasing C raises the penalty for margin violations → typically tighter fit / lower regularization. Increasing γ makes the RBF influence more local → more intricate boundaries. Underfit: tiny C + tiny γ. Overfit: very large C + very large γ on noisy data.

4. Gini gain

Parent G=0.5. Split [18,2]/[2,18]: each child G=0.18, weighted G=0.18, gain=0.32. Split [12,8]/[8,12]: each child $G=1-(0.6^2+0.4^2)=0.48$, weighted G=0.48, gain=0.02. First split is far better.

INSTRUCTOR APPENDIX

HW 5–6 solutions

5. Overfitting tree

A depth-20 tree can split on room/microphone/noise details that correlate with labels only in the training set. Use max_depth, min_samples_leaf, max_leaf_nodes, or ccp_alpha; validate choices by CV.

6. Random forest

Each tree sees a bootstrap sample; each split sees a random subset of features. OOB prediction uses trees for which a sample was not in the bootstrap training set. Averaging reduces variance mainly when tree errors are not perfectly correlated; correlated errors survive averaging.

INSTRUCTOR APPENDIX

HW 7–8 solutions / grading guidance

7. Notebook comparison

Accept the student's reproducible 5-fold CV numbers. Full-credit interpretation: PCA is optional; compare mean and variability, and do not claim superiority from a tiny difference inside CV uncertainty.

8. Importance with correlated duplicate

Expected: importance can be divided between correlated features; MDI and permutation may respond differently. A duplicate can reduce the apparent permutation importance of either feature because the other still supplies similar information.

Grading emphasis: reasoning > matching a specific numerical importance ranking.

INSTRUCTOR APPENDIX

Weekly quiz solutions

Answers

1. B — variance. 2. C — avoid leakage. 3. B — stronger penalty on margin violations. 4. C — decision tree. 5. B — de-correlate trees.

6. Short answer

Feature importance measures predictive usefulness to this fitted model/data distribution. Correlation, confounding, data collection bias, or a proxy variable can make a feature predictive without being a physical cause.

Common misconception to correct

“Important = causal” and “PCA = supervised separation” are both false.

INSTRUCTOR RESOURCES

Files supplied with this week

PowerPoint

Week12_DSP_ML_PCA_SVM_Trees_RF.pptx — both 75-minute lectures + assessment + instructor solution appendix.

Lecture 23 notebook

Lecture23_PCA_SVM.ipynb — synthetic audio features, PCA, scree/loadings, linear & RBF SVM, GridSearchCV, leakage-safe pipeline.

Lecture 24 notebook

Lecture24_Trees_RandomForests.ipynb — impurity, tree visualization, overfitting/pruning, forest/OOB, feature importance, CV comparison.